Homework — 1.1.1 Structure and Function of the Processor — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [3] — 1 mark each (full name **and** role both required): - PC = **Program Counter** — holds the address of the **next** instruction to be fetched (incremented each cycle). - ACC = **Accumulator** — stores the **immediate result** of ALU calculations. - MAR = **Memory Address Register** — holds the **address** of the location to be read from / written to.

2. [2] — 1 mark per point (max 2): decodes instructions; generates control signals; manages/directs the flow of data on the buses; synchronises operations with the clock.

3. [3] — 1 mark each, in order: - A **read signal** is sent on the control bus; the instruction returns via the **data bus into the MDR**. - The **PC is incremented** (PC := PC + 1). - The contents of the **MDR are copied into the CIR**. *(Accept increment of PC and MDR→CIR in either of the last two positions.)*

4. [2] - (a) **[1]** — 2^{20} = **1,048,576 locations**. *(Accept "1M / 1048576".)* - (b) **[1]** — A wider data bus carries **more bits per transfer**, increasing throughput / fetching more data per cycle (faster data movement).

5. [2] — 1 mark: each core has its own FDE cycle, so multiple instructions/tasks can be processed **simultaneously (in parallel)**. 1 mark: only if the **task can be parallelised AND the software is written to use multiple cores/threads**.

6. [2] — 1 mark: **opcode** — specifies the operation to be carried out. 1 mark: **operand** — the data or the address of the data to operate on.

Part B (6 marks)

7. [3] — 1 mark each: - Address bus = **unidirectional** (CPU → memory), carries the **memory address**. - Data bus = **bidirectional**, carries **data/instructions**. - Control bus = **bidirectional**, carries **control signals** (e.g. read/write, clock, interrupt).

8. [2] — 1 mark: cache is **small, very fast memory on/near the CPU** holding frequently/recently used data/instructions (not part of RAM). 1 mark: it reduces the need to fetch from **slower RAM**, so data is retrieved faster / fewer slow main-memory accesses.

9. [1] — **L3** is larger (L1 is the smallest and fastest).

Total: 14 + 6 = 20